Session 1: AI for Coding — Tools, Concepts, and Setup

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What We Will Cover (Sessions 1 & 2)

- Part 1: Tools, concepts, and setup
  - Overview of AI coding tools and how to choose among them
  - How to use AI Agents: Terminal, Desktop App, VS Code
  - What Git is and why it matters for research
  - Python coding with AI

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- Part 1: Tools, concepts, and setup
  - Overview of AI coding tools and how to choose among them
  - How to use AI Agents: Terminal, Desktop App, VS Code
  - What Git is and why it matters for research
  - Python coding with AI
- Part 2: Practical workflows
  - Writing, debugging, and automating code with AI
  - “Vibe coding” from plain English descriptions
  - Project organization and replication packages
  - Demo: data cleaning script from scratch

Coding Perhaps Highest-Value Use of AI

- AI most reliable when output is verifiable: code runs or it does not
- Economists spend enormous time on repetitive coding tasks
  - AI can handle most of them
- Massive productivity gains across skill levels:
  - Beginners: “vibe code” to write code in an unfamiliar language (no syntax!)
  - Advanced users: automate drudge work, prototype faster, reduce bugs

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$\Rightarrow$ First two sessions will cover how to use AI to complete coding tasks

Overview of AI Coding Tools

The AI Coding Landscape: Many Options

- General-purpose AI assistants (good at coding, among many other things):
  - **Claude** (Anthropic) — our focus today
  - **ChatGPT / GPT-4o** (OpenAI)
  - **Gemini** (Google)
- Agentic coding tools (read files, write and run code autonomously):
  - **Claude Code** (Anthropic) — our focus today
  - **Codex** (OpenAI) — closest competitor to Claude Code
  - **GitHub Copilot** (Microsoft) — inline suggestions in editor
  - **Cursor / Windsurf** — AI-first code editors
- How to choose: task type, privacy needs, and workflow integration

Chatbot vs. Agentic Tool

- **Chatbot** (Claude.ai, ChatGPT): you ask → it answers → *you* act
- **Agentic tool** (Claude Code): you give a goal → it reads files, writes code, runs it, fixes errors, iterates → reports back
- For economists: instead of "here is some code," it runs the code and tells you what it found

Which One Should You Use? Claude Code vs. Codex

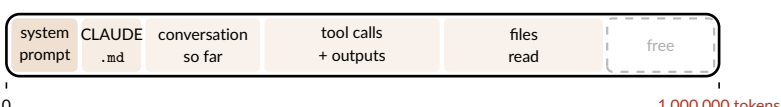
- Two serious options for agentic coding today: **Claude Code** and **Codex**
- Plans mirror each other at each price point:
  - $20/mo: Claude Pro vs. ChatGPT Plus
  - $100/mo: Claude Max (5x) vs. ChatGPT Pro (5x)
  - $200/mo: Claude Max (20x) vs. ChatGPT Pro (20x)
- Both use a 5-hour rolling window + weekly cap; limits reset automatically
- At the same price, Codex currently gives more usage per dollar

Which One Should You Use? Claude Code vs. Codex

- Two serious options for agentic coding today: **Claude Code** and **Codex**
- **Codex** has some advantages:
  - More usage per subscription dollar — important for heavy users
  - Sometimes stronger on pure coding benchmarks
- **Claude Code** has some advantages:
  - Often better at tasks beyond code: understanding context, writing, reasoning
- **Bottom line:** this comparison is fast-moving and will keep changing
  - I am using Claude Code and find it excellent for my workflow
  - I would switch only if Codex proved clearly and consistently superior
  - My advice: pick one, learn it well, reassess in 6 months

What You Pay For: Tokens and Context

- **Token**: the unit AI reads and writes — roughly $\frac{3}{4}$ of a word
  - **You pay for**: tokens *in* (prompt + files) and tokens *out* (reply) — input is cheaper
- **Context window**: everything the model holds at once — fills as session grows



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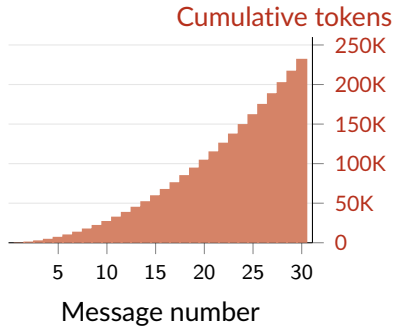

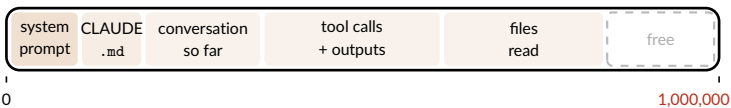

Getting the Most Out of Your Subscription

- Subscriptions have rolling limits — fewer tokens per task means more use
- How to use fewer tokens:
  - Start a new session for each new task: context accumulates and raises costs
  - Store standing instructions once so you never repeat yourself
  - Be specific: vague prompts waste tokens and invite unfocused replies
  - Ask multiple things at once so less back and forth
  - Compact your chats/context window
- Silly trick: start Claude when wake up so 5-hour window ends earlier in day

Claude Code / Codex Interfaces

Three Ways to Use an Agentic Coding Tool

- **Terminal** (command line):
  - Most powerful; works directly in your project directory
  - Best for: running scripts, data analysis, replication tasks
- **Desktop App**:
  - Friendlier interface; easier to get started
  - Projects feature: persistent context across sessions
  - Best for: drafting, quick questions, early exploration
- **VS Code integration**:
  - Agent lives inside your editor — see code and AI side by side
  - Best for: sustained development work, Git integration, Jupyter/Python

Terminal vs. VS Code: Quick Comparison

Terminal (CLI)

- Lightweight; fast startup
- Works anywhere, incl. remote servers
- Keyboard-driven; fast for quick tasks
- No inline diffs; steeper setup

VS Code Extension

- Inline diffs; accept changes block-by-block
- Code & AI panel side by side
- Best for large codebases, sustained work
- Heavier on CPU and memory

Live Demo: Terminal

- ▷ *Demo slide — minimal text, walk through live*
- Navigate to a project directory
- Ask Claude Code to describe what it sees
- Ask it to write a simple data task

Live Demo: Desktop App

- ▷ *Demo slide — minimal text, walk through live*
- Create a Project and load context
- Same task as terminal — compare experience

Live Demo: VS Code

- ▷ *Demo slide — walk through VS Code interface live*
- Same task as terminal — compare experience

CLAUDE.md and Skills

- **CLAUDE.md** (memory) — Claude reads at the start of every session
  - Store context that is always true so you don't repeat yourself in every prompt
  - `.claude/CLAUDE.md` is global, `your-project/CLAUDE.md` is *project* specific
  - Ask Claude to update after substantial changes to project
- **Skills** — Claude reads every time it is called
  - Store context for a task that you repeat often, but not always relevant
  - `.claude/skills/your-skill/SKILL.md` is called with `/your-skill`
  - Write and update your own skills or download from others

Read more here (full descriptions, examples of CLAUDE.md and skills, download pre-made skills)

`your-project/CLAUDE.md` (example from Claes Backman's Guide)

```
# Project: [Paper title]

## Data
- Unit of analysis: firm-year panel, 2010–2022
- Raw data lives in data/raw/ — never edit these files
- Main dataset after cleaning: data/clean/panel.dta (N ≈ 47,000)

## Code conventions
- R scripts are numbered (01_clean.R, 02_analysis.R, ...)
- All regressions cluster SEs at the firm level
- Use fixest for all panel regressions

## LaTeX
- Main file: paper/main.tex
- Tables generated by 03_tables.R go to paper/tables/
- Use \widehat{} not \hat{} for estimators
```

`.claude/skills/robustness/SKILL.md` (example from Claes Backman's Guide)

```
---
name: robustness
description: Run standard robustness checks on the main regression.
---

For the main specification in this project:
1. Re-run with alternative clustering (industry-year instead of firm)
2. Re-run dropping the top and bottom 1% of the outcome variable
3. Re-run on the pre-2020 subsample only
4. Produce a summary table comparing coefficients across all specs
```

VS Code with Claude Overview

`.claude/skills/robustness/SKILL.md` (example from Claes Backman's Guide)

- Key panels: Explorer (files), Editor, Terminal, Source Control (Git), Extensions, Claude Code
- CC Modes:
  - **Ask before edits**: Claude proposes each change; you approve
  - **Plan mode**: Claude writes a full plan in Markdown before doing anything
  - **Auto**: Claude acts autonomously (use carefully)

Git: Version Control for Economists

Why Git Matters for Research

- Version control = a complete history of every change to your code
- No more: `analysis_v3_FINAL_revised2.do`
- Enables:
  - Rolling back to any prior version of your code
  - Working safely with coauthors and RAs without overwriting each other
  - Tracking exactly what changed between draft versions
  - Automatic replication audit trail
- The good news: Claude Code can handle all Git commands for you — you just describe what you want

Key Git Concepts (No Commands Required)

- **Repository (repo)**: a folder whose history Git is tracking
- **Commit**: a snapshot of your code at a point in time, with a message
- **Push / Pull**: sync your local repo with GitHub (the remote copy)
- **Branch**: a parallel version of the code for a new feature or robustness check
- **Merge**: fold a branch back into the main code
- Typical workflow for an economist:
  1. Work on code; when something works, commit it
  2. Push to GitHub at end of day (backup + coauthor access)
  3. RA works on a branch; you review and merge

Git and Claude Code: What the Integration Looks Like

- ▷ *Placeholder: screenshot or demo*
- In VS Code: Git panel shows all changes; Claude Code can stage, commit, push on request
- You tell Claude: “commit my changes with a descriptive message” — it writes the message too
- Branching for research: create a branch for each robustness check, merge if it works
- Connecting with Overleaf:
  - Overleaf Pro can sync with a GitHub repo
  - Alternatively: LaTeX Workshop extension in VS Code + local compilation

Python in VS Code: Environments

- ▷ *Demo slide — Tanner*
- Anaconda manages Python installations and packages
- Each project gets its own **environment**: a fixed set of packages and versions
  - Prevents conflicts between projects
  - Ensures others can reproduce your results exactly
- Ask Claude Code to create and activate an environment for you

Running Python Code with Jupyter

- ▷ *Demo slide — Tanner*
- Three modes in VS Code:
  - **Jupyter notebook** (`.ipynb`): interactive cells, inline output — good for exploration
  - **Interactive window**: run `.py` scripts cell-by-cell — best of both worlds
  - **Script** (`.py`): run the full pipeline — best for production code
- Claude Code can write, run, and debug Python in any of these modes

Setting Up Your Environment

What You Need to Install

- Core tools (all free except AI subscription):
  1. **VS Code** — the editor (free)
  2. **Claude Code** — VS Code extension + CLI (requires subscription)
  3. **Git** — version control (free); create a GitHub account
  4. **Python** — Anaconda distribution recommended (free)
- VS Code extensions to install:
  - Claude Code, GitLens, Jupyter, Python, Stata Enhanced (if applicable)
- We have a step-by-step setup guide — see handout / shared folder

Session 1 — Key Takeaways

1. AI has perhaps most value added in coding
2. Agentic $\neq$ chatbot: reads files, writes and runs code, fixes errors, reports back
3. Two serious agentic options today: Claude Code and Codex
   - Pick one, learn it, reassess in 6 months
4. Tokens accumulate fast in long session — start new session for each new task
5. Three interfaces (Terminal, Desktop, VS Code) and Git are your essential stack

Questions?

- Questions on tools, concepts, or setup?
- Thoughts on how this fits your current workflow?